

COMPONENTS

Automotive speech synthesis IC

LAPIS Semiconductor has introduced the ML2253x series of automotive speech synthesis ICs that have been optimised for audible notifications and sound effects in the advanced driver assistance system (ADAS) and acoustic vehicle alerting system (AVAS). The new product series not only improves quality by detecting audio defects using a playback anomaly function but also facilitates system configuration while simplifying software design for the main MCU.

The ML2253x series also includes a playback sound anomaly function that can send error signals to the main MCU. This makes it possible to detect playback anomalies such as skipping, thereby further improving the quality of automotive audio output systems.

Compared with a conventional compression algorithm, HQ-ADPCM enables to achieve high fidelity sound without distortion. Additionally, flash rewrite function from the main MCU enables rewriting of voice data even after shipment.

LAPIS Semiconductor
www.lapis-semi.com/en/

Imaging chip

RFISee, an Israel-based developer of imaging radars for the automotive industry, has launched phased array 4D imaging radar on a chip. RFISee's all-weather radar can detect cars from 500m and pedestrians from 200m, with an angular resolution greater than one degree. The phased array antenna technology is used in cutting edge military systems, including F-35 fighter jets and in air defence systems.

The company's 4D imaging radar uses a powerfully focused beam based

on proprietary phased array radar technology. The focused beam created by dozens of transmitters rapidly scans the field of view. The receivers ensure a much-improved radar image, a better signal to noise ratio and a wide detection range (nearly 6x compared to existing radars) of obstacles such as cars and pedestrians.



The radar's features prevent accidents that go undetected by other existing radar systems.

RFISee
www.rfisee.com

MEMS speaker

xMEMS Labs, a developer of speaker technology for personal audio devices, has introduced Montara, a MEMS speaker that can deliver high-fidelity, full-bandwidth sound and low total harmonic distortion (THD) for sealed in-ear personal audio devices, including true wireless stereo (TWS) earbuds.

Montara is an IP-57 rated micro-speaker, enabling water and dust-resistant earbuds without special protective mesh or coverings. Consumers can benefit from enhanced audio fidelity with raised levels of clarity, extremely low latency, waterproof earbuds, and longer listening times.

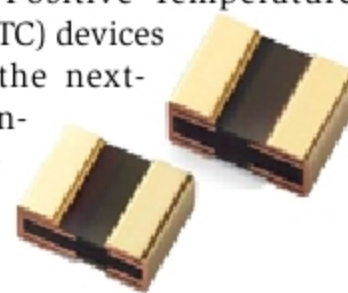


Montara replaces the previous highly-variable voice-coil speaker technology with a high-precision, high-consistency, monolithic semiconductor process, thus reducing package height and eliminating driver matching and calibration due to inherent variability in membrane assembly.

xMEMS Labs
www.xmems.com

PPTC devices

Littelfuse has launched a new series of Polymeric Positive Temperature Coefficient (PPTC) devices designed for the next-generation consumer electronics to become smaller and more reliable.



The new series provides robust overcurrent and overvoltage protection on the signal lines for both li-ion battery pack ICs and fuel gauge. These PPTC devices are for high voltage (13V DC) applications and ideal for designing increasingly small mobile and wearable product designs such as smartphones, notebooks, etc.

The zeptoSMDC series PPTC devices are available in small 0201 surface-mount sizes in tape and reel format.

Littlefuse
www.littelfuse.com

Power amplifier

Qorvo has introduced a GaN power amplifier (PA) that delivers a fifty per cent increase in power for improved range, performance, and multi-target tracking in S-band (2-4GHz) phased array radars.

With 150 watts of power for the 2.9 to 3.5GHz frequency range, the QPA3070 achieves 58 per cent power added efficiency (PAE) and 28dB power gain with the help of ultra-reliable and highly efficient gallium nitride (GaN)

